

DTS 201: Introduction to Data Science

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Real life Examples

01

Your bank flagged a transfer

A model compared it against thousands of your past transactions and scored it as unusual.

02

Maps rerouted you around Lekki

Live GPS traces from other phones predicted the hold-up before you ever reached it.

03

Your feed reordered itself

Ranking models chose which of thousands of posts you were most likely to stop scrolling for.

04

A ride was priced just for you

Demand, supply, weather and time of day produced a fare that did not exist an hour ago.

05

Spam emails are identified

A classifier trained on billions of messages filed it away without ever asking you.

06

You get personalised ads

An e-commerce sees the type of products you add to your cart or wishlist and it begins to notify you of other products you might like that are in stock.

Course Outline

WK	TOPIC	WK	TOPIC
01	Introduction to Data Science	08	Data Pre-processing & Cleaning
02	Data, Data Types & Data Sources	09	Machine Learning I: Foundations
03	Data Collection, Sampling & Quality	10	Machine Learning II: Models & Evaluation
04	Descriptive Statistics & Distributions	11	Data Mining (Clustering & Anomaly Detection)
05	Probability & Simulation	12	Data Science Across Domains
06	Analytics, Knowledge Extraction & Visualization	13	Summary Class & Take-Home Review
07	Summary Class & First Test	14	Big Data, BI & Data Science in Practice

Introduction to Data Science

Learning Objectives

- Understand what data science entails
- The data science lifecycle
- Applications of data science
- Get familiar with the Google's cloud-hosted Jupyter Notebook

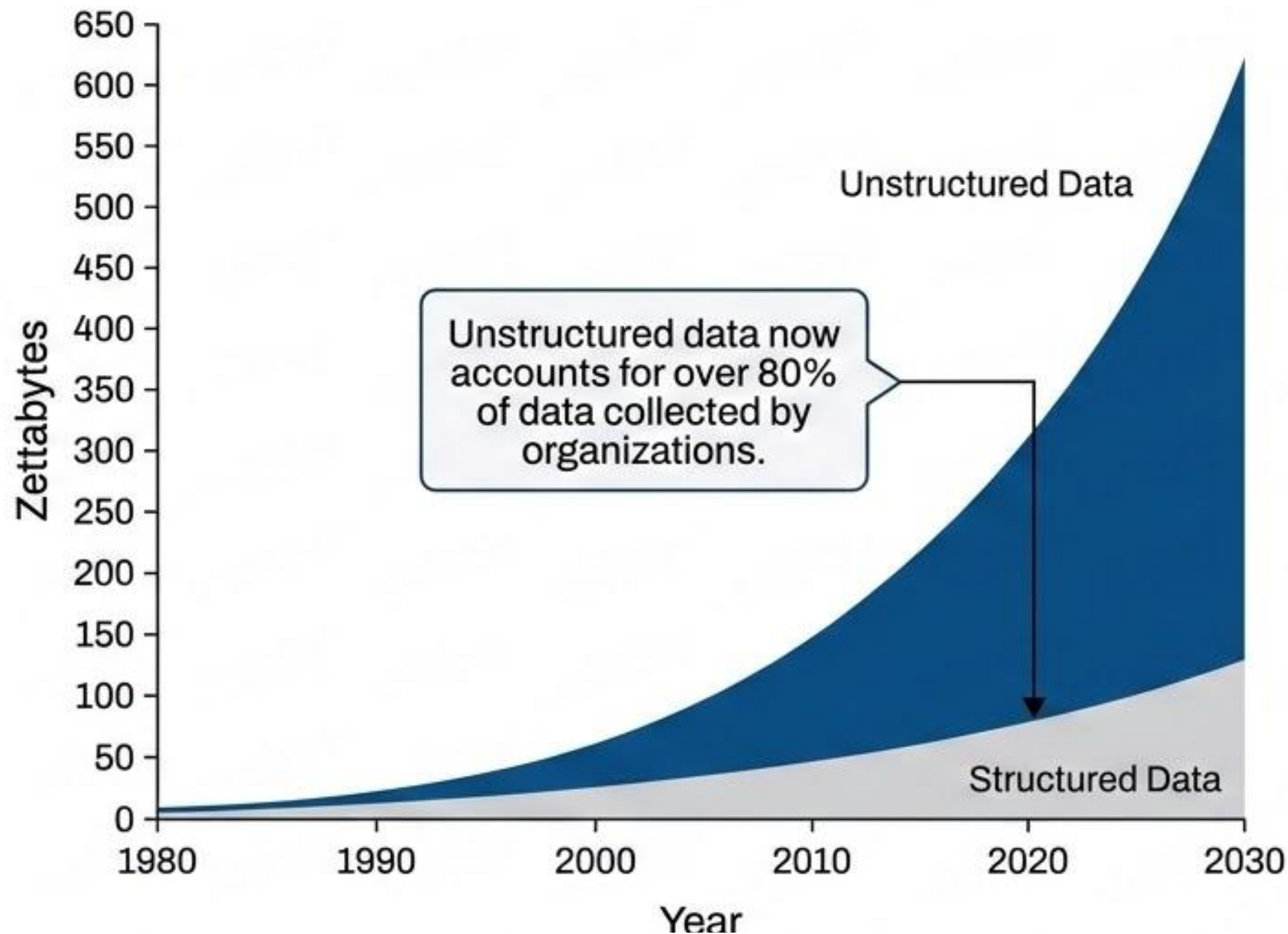
What is data?

Data is a recorded observation about the world.

This could be a measurement written down, a transaction detail, a photograph, a voice recording, your chat prompts, blog posts, Uber ride location, etc.

Data becomes useful only in **quantity** and in **context** because this allows you to compare it against other observations, only then can you make sense of it.

The Zettabyte Era: Why Data Science is Essential



181 Zettabytes

Estimated global data volume generated in 2024.

2.0 Billion+

Active daily users on a major social platform.

612 Zettabytes

Projected global data volume by 2030.

What is Data Science?

The primary goal of data science is to uncover hidden patterns, generate actionable insights and build predictive models to help humans (or businesses) make data-driven decisions.

For example, how can a data scientist help farmers:

Data

Say you have 3,000 daily rainfall readings from across the state, in the last 10 years

Information

You discover that the data shows that in this year, rain began about two weeks later than it did a decade ago.

Knowledge

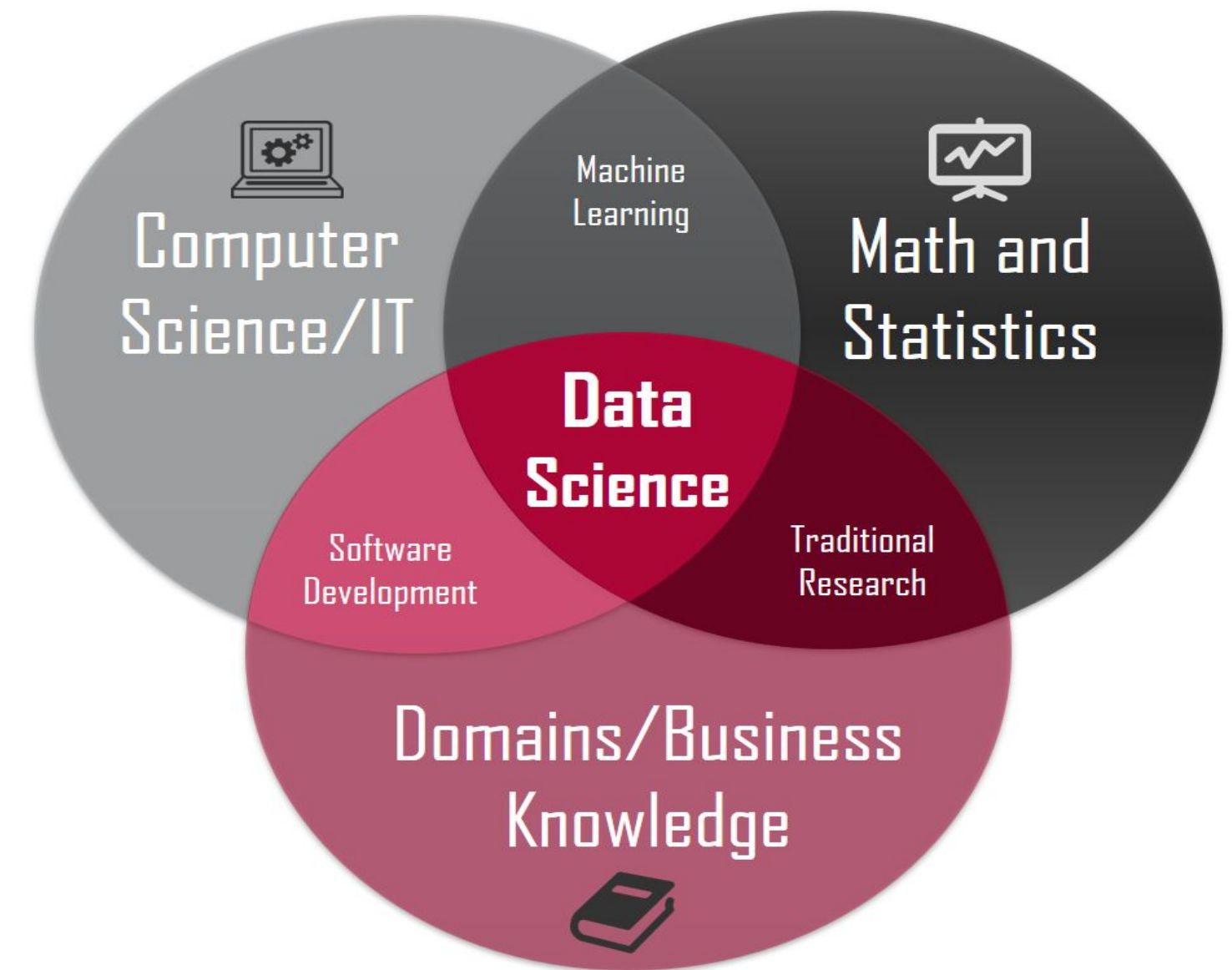
This informs you that planting on the old calendar may risk losing the early crop.

Decision

Advise farmers to shift planting to a future date.

What is Data Science?

Data science is a multidisciplinary field that combine statistics & mathematics, programming, and domain knowledge to clean, process, analyse and interpret large amounts of data.



Discipline	Purpose in data science	Example
Statistics	Core foundation for quantifying data, testing hypotheses, and making valid inferences from data.	Evaluating deviations from mean, calculating confidence intervals.
Mathematics	Provides the theoretical understanding (linear algebra, calculus, probability) of how machine learning algorithms work.	Vector and matrix computations in neural networks, gradient descent optimization.
Programming	The practical toolset used to ingest, clean, manipulate, and datasets, also used for building and deploying models.	Writing Python or R scripts to automate ETL pipelines and train predictive models.
Domain Knowledge	Crucial for asking the right questions, designing meaningful features, and translating data insights into action.	Understanding key business metrics (like churn or LTV) to guide model development.

Why data science is important to businesses

- **Provides more certainty in decision making:** Making decisions when you can confidently back up the “why” is very important in a business environment because they can track what worked and what didn’t rather than relying on intuition, which could be subjective.
- **It saves resources:** Data science can help a business reveal a case that is worth flagging or a customer that is worth a discount, therefore saving the resources that could have been spent on targeting the wrong audience.
- **Makes a business proactive:** It can help a business identify what is likely to happen or what a user may respond to and make early decisions, rather than reacting, which could be fatal in some cases.
- **Risk Mitigation:** With data science, businesses can identify anomalies and fraudulent patterns, therefore reducing the risks that occur in certain business operations.
- **Improved customer experience:** Overall, when a business begin to make intelligent decisions, the experience of the customers (whether internal or external) become improved.

The data science lifecycle

The Seven Stages in a Data Science Cycle



Stages 2, 3 and 4 routinely consume most of a project's time.

As stated, it is a cycle

You will go back and forth between each stage.

The lifecycle is drawn as seven neat boxes because it is easier to teach that way. In practice it loops, and each loop sharpens the previous or next questions.

For example, expect to redefine the problem after seeing the data. Expect to collect more data after exploring. Expect evaluation to send you back to modelling.

Exploration → Problem

“This data cannot answer that question, what do we do?”

Preparation → Collection

“Half the records are missing income. We need another source.”

Evaluation → Modelling

“The performance is not so good, how can we improve it?”

Monitoring → The Beginning

“It is worked last year. The industry has changed.
Start again.”

Applications of data science

Finance & healthcare

Finance

Banks and fintechs run on models. Every transfer you make is scored in milliseconds.

Fraud detection on live transaction streams

Credit scoring for customers with no formal credit history

Algorithmic trading and risk modelling

Predicting which customers are about to leave

Healthcare

Where diagnostic specialists are scarce, models extend the reach of the few who exist.

Reading X-rays and scans to flag likely cases

Tracking outbreaks from clinic and search data

Predicting patient readmission risk

Narrowing drug candidates before laboratory testing

Agriculture & education

Agriculture

Satellite imagery, soil sensors and weather records turned into advice a farmer can act on this week.

Yield forecasting from satellite and rainfall data

Detecting crop disease from leaf photographs

Advising planting dates as rainfall patterns shift

Credit scoring smallholders using farm data

Education

The sector you are sitting in right now – and one where the data is unusually rich.

Early identification of students at risk of dropping out

Adaptive learning platforms that adjust to each learner

Analysing which teaching methods actually work

Planning admissions, timetables and resources

Telecommunications & government

Telecommunications

Among the largest datasets anywhere: every call, every top-up, every tower connection.

Predicting which subscribers will switch networks

Deciding where to place the next mast

Detecting SIM fraud and unusual usage

Designing data bundles from real usage patterns

Government

Public policy that is measured rather than assumed — with real questions about privacy attached.

Census, planning and resource allocation

Tax compliance and revenue leakage detection

Traffic, transport and urban planning

Targeting social intervention programmes

Retail & social media

Retail & e-commerce

Activities such as scrolling, clicking, buying or even abandoning a cart is data point (observation).

Recommendation engines “customers also bought”

Demand forecasting and inventory planning

Dynamic pricing and promotion targeting

Routing deliveries across congested cities

Social media

The sector where data science most visibly shapes what a whole society sees and believes.

Ranking what appears in your feed

Sentiment analysis of public opinion

Detecting misinformation and coordinated accounts

Ads recommendation

Python for data science



Notebook Lab Session with Google Colab

A notebook is a document where text and runnable code live side by side. This is perfect for data analysis.

- 01 Go to **colab.research.google.com** and sign in with a Google account.
- 02 Create a **New notebook**. Rename it to *DTS201-Week1*.
- 03 Type something into the first cell and press **Shift + Enter** to run it.
- 04 That is it. Your notebook saves to Google Drive automatically.

We'll practicalize the following:

- Understanding how to use the notebook
- Exploring variables and Data types (int, float, str, bool, dictionary, list)
- Calculations
- Built-in functions (sum, len, min, max) and importing libraries (for now “statistics”)

Class exercise

In your Colab notebook, work with this small dataset:

```
ages = [19, 22, 21, 35, 28, 24, 31, 19, 27, 40]
```

- a. How many people are in the survey?
- b. What is the total of all the ages?
- c. What is the mean age? Print it to one decimal place.
- d. What are the youngest and oldest ages?
- e. Import `statistics` and find the median. Is it different from the mean?
When the mean and the median disagree, the data is telling you something. What?

Check in

Q1 In one sentence: what is data science?

Q2 Name the seven stages of the data science lifecycle, in order.

Q3 Which stage usually consumes the most time, and why?

Q4 Name a data science application in a sector you care about.
